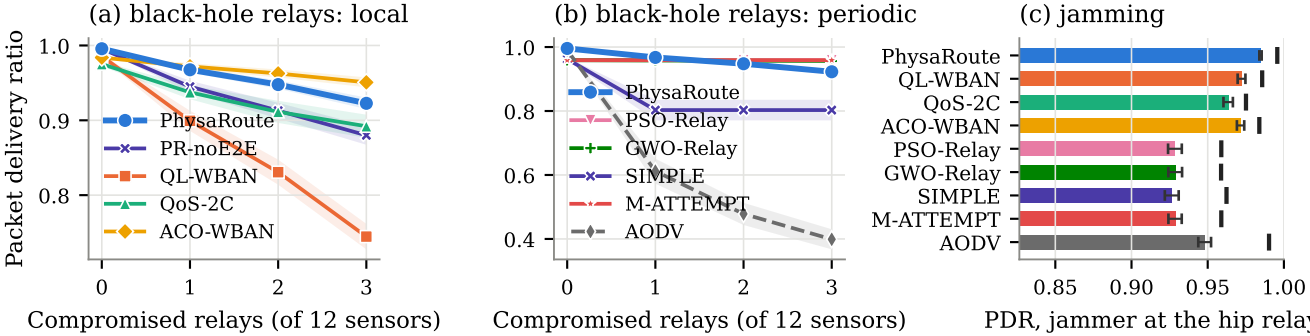
(a) black-hole relays: local
Packet delivery ratio
1.0
0.9
0.8
PhysaRoute
PR-noE2E
QL-WBAN
QoS-2C
ACO-WBAN
0
1
2
3
Compromised relays (of 12 sensors)
(b) black-hole relays: periodic
1.0
0.8
0.6
0.4
PhysaRoute
PSO-Relay
GWO-Relay
SIMPLE
M-ATTEMPT
AODV
0
1
2
3
Compromised relays (of 12 sensors)
(c) jamming
PhysaRoute
QL-WBAN
QoS-2C
ACO-WBAN
PSO-Relay
GWO-Relay
SIMPLE
M-ATTEMPT
AODV
0.85
0.90
0.95
1.00
PDR, jammer at the hip relay
no jammer